

TASK 1: DNA/Protein Sequence Analysis Using BLAST

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Title -

Identification and Homology Analysis of Hemoglobin Beta Protein Using BLASTP

Aim

To analyze a protein sequence obtained from a biological sequence database and identify homologous proteins using the NCBI BLASTP tool. The analysis focuses on similarity, percentage identity, alignment score, query coverage, and evolutionary conservation..

Introduction

BLAST (Basic Local Alignment Search Tool) is a widely used bioinformatics method for comparing a query sequence with sequences in biological databases. For protein sequences, BLASTP identifies proteins with significant sequence similarity and provides measures such as percentage identity, query coverage, E-value, bit score, and sequence alignment. In this study, the human hemoglobin beta-chain sequence was analyzed.

The screenshot shows the NCBI Protein database record for human hemoglobin subunit beta (NP_000509.1). The page includes the following information:

- Protein:** hemoglobin subunit beta [Homo sapiens]
- NCBI Reference Sequence:** NP_000509.1
- Identical Proteins:** FASTA Graphics
- Go to:** [Link]
- LOCUS:** NP_000509 147 aa linear PRI 14-JUN-2026
- DEFINITION:** hemoglobin subunit beta [Homo sapiens].
- ACCESSION:** NP_000509
- VERSION:** NP_000509.1
- DBSOURCE:** RefSeq; accession [NM_000518.5](#)
- KEYWORDS:** RefSeq; NAME Select.
- SOURCE:** Homo sapiens (human)
- ORGANISM:** [Homo sapiens](#)
Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Hominidae; Homo.
- REFERENCE:** 1 (residues 1 to 147)
- AUTHORS:** Civettini, I., Zappaterra, A., Corti, P., Messina, A., Aroldi, A., Biondi, A., Cavalca, F., Crippa, V., Crosti, F., Ferrari, G.M., Malighetti, F., Mologni, L., Piperno, A., Ramazzotti, D., Scollo, C., Zambon, A., Rossi, F. and Gambacorti-Passerini, C.
- TITLE:** Hb Nonca: A novel extensive HBB duplication with preserved alpha-beta subunit interaction and unstable hemoglobin phenotype
- JOURNAL:** Med 6 (4), 100554 (2025)
- Protein 3D Structure:** CD163 bound to hemoglobin PDB: 9TQD Source: Homo sapiens Method: Electron Microscopy Resolution: 2.8 Å

Figure 1. NCBI Protein database record for human hemoglobin subunit beta (NP_000509.1), showing the protein sequence information and accession details.

For this analysis, a **147-amino-acid protein sequence** was analyzed using NCBI BLASTP. The search was performed against the **NCBI non-redundant (nr) protein database**. The BLAST results were examined to identify the most similar proteins and determine the identity of the query sequence.

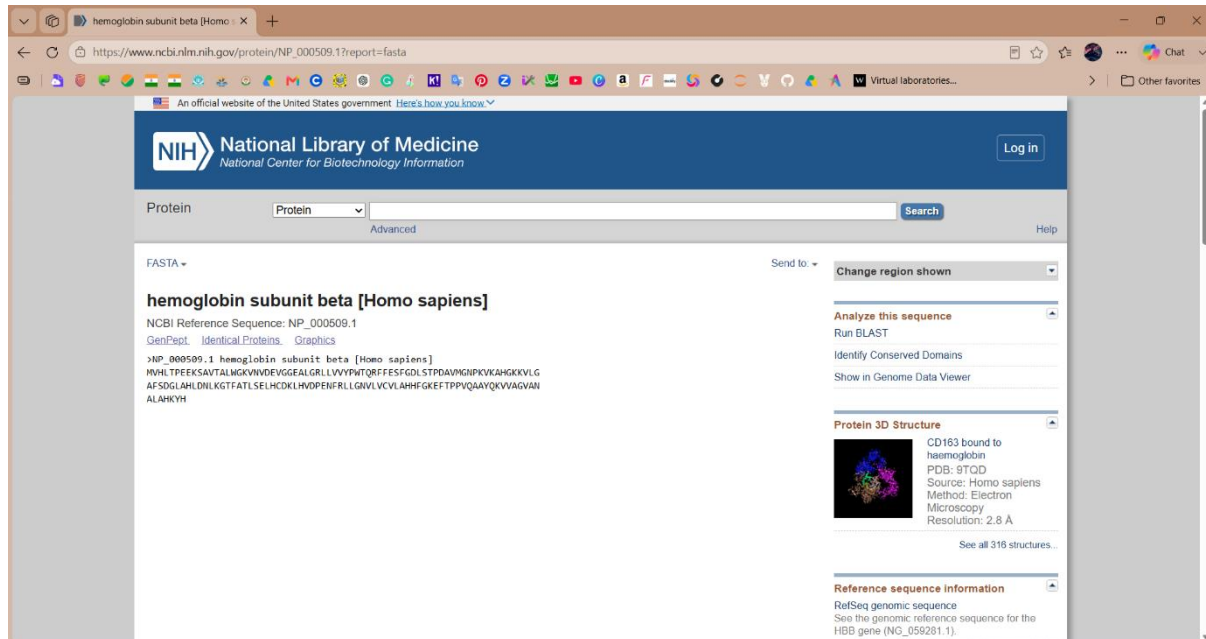


Figure 2. FASTA-formatted protein sequence of human hemoglobin subunit beta retrieved from NCBI and used as the input sequence for BLASTP analysis

Materials and Methods

Sequence Selection

A protein sequence of **147 amino acids** was selected for analysis. The sequence was analyzed using the NCBI BLASTP program.

BLASTP Analysis

The following procedure was followed:

1. The protein sequence was entered into the NCBI BLASTP search tool.
2. The **nr (non-redundant protein)** database was selected.

Protein BLAST: search protein data

https://blast.ncbi.nlm.nih.gov/Blast.cgi?PROGRAM=blastp&PAGE_TYPE=BlastSearch&LINK_LOC=blasthome

NIH National Library of Medicine
National Center for Biotechnology Information

Log in

BLAST® » blastp suite

Standard Protein BLAST

blastn blastp blastx tblastn tblastx

BLASTP programs search protein databases using a protein query, more...

Enter Query Sequence

Enter accession number(s), g(i), or FASTA sequence(s) Clear

Query subrange From To

Or, upload file Choose File No file chosen

Job Title

Align two or more sequences

Choose Search Set

Database Non-redundant protein sequences (nr)

Organism Optional

Exclude Optional

Program Selection

Algorithm

Quick BLASTP (Accelerated protein-protein BLAST)

blastp (protein-protein BLAST)

PSI-BLAST (Position-Specific Iterated BLAST)

PI-BLAST (Pattern Hit Initiated BLAST)

DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm

BLAST

Search database nr using blastp (protein-protein BLAST)

Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with a sign.

Figure 3. NCBI BLASTP search setup showing the entered protein sequence and selected non-redundant protein database (nr)

3. The BLASTP search was submitted.
4. The resulting homologous protein sequences were examined.
5. The top matches were compared based on:
 - Percentage identity
 - Query coverage
 - E-value
 - Bit score
 - Number of identical and positive residues
 - Number of gaps
6. The alignment between the query sequence and the best matching proteins was examined.

NCBI BlastProtein Sequence

https://blast.ncbi.nlm.nih.gov/Blast.cgi

NIH National Library of Medicine
National Center for Biotechnology Information

Log in

BLAST® » blastp suite » results for RID-8NH7N.J4P014

Home Recent Results Saved Strategies Help

How to read this report? BLAST Help Videos Back to Traditional Results Page

Job Title Protein Sequence

RID 8NH7N.J4P014 Search expires on 09-24 18:01 pm Download All

Program BLASTP Citation

Database nr See details

Query ID IcdQuery_3919622

Description unnamed protein product

Molecule type amino acid

Query Length 147

Other reports Distance tree of results Multiple alignment MSA viewer

Filter Results

Organism only top 20 will appear

Type common name, binomial, taxid or group name

Add organism

Percent Identity E value Query Coverage

Filter Reset

Compare these results against the new Clustered nr database BLAST

Descriptions Graphic Summary Alignments Taxonomy

Sequences producing significant alignments

Download Select columns Show 100

select all 100 sequences selected

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	hemoglobin, beta, partial [synthetic construct]	synthetic const...	301	301	100%	1e-102	100.00%	148	AA337851.1
☒	hemoglobin, beta, partial [synthetic construct]	synthetic const...	301	301	100%	1e-102	100.00%	148	AA329567.1
☒	hemoglobin subunit beta [Homo sapiens]	Homo sapiens	301	301	100%	1e-102	100.00%	147	NP_009509.1
☒	hemoglobin subunit beta [Gorilla gorilla gorilla]	Gorilla gorilla g...	300	300	100%	5e-102	99.32%	147	XP_018891789.1

Results

The BLASTP search produced several highly significant matches. The strongest biologically relevant match was **hemoglobin subunit beta [Homo sapiens]**, with accession number **NP_000509.1**.

The query sequence and the human hemoglobin beta sequence were both 147 amino acids long. The complete query sequence aligned with the reference sequence.

Table 1. BLASTP result for the best match

Parameter	Result
Query protein length	147 amino acids
Database	NCBI nr protein database
BLAST program	BLASTP
Best match	Hemoglobin subunit beta [Homo sapiens]
Accession number	NP_000509.1
Query coverage	100%
Sequence identity	147/147 (100%)
Positives	147/147 (100%)
Gaps	0/147 (0%)
E-value	1e-102
Bit score	301 bits

The alignment showed that all 147 amino acids of the query sequence were identical to the corresponding amino acids in the human hemoglobin beta protein. There were no gaps in the alignment.

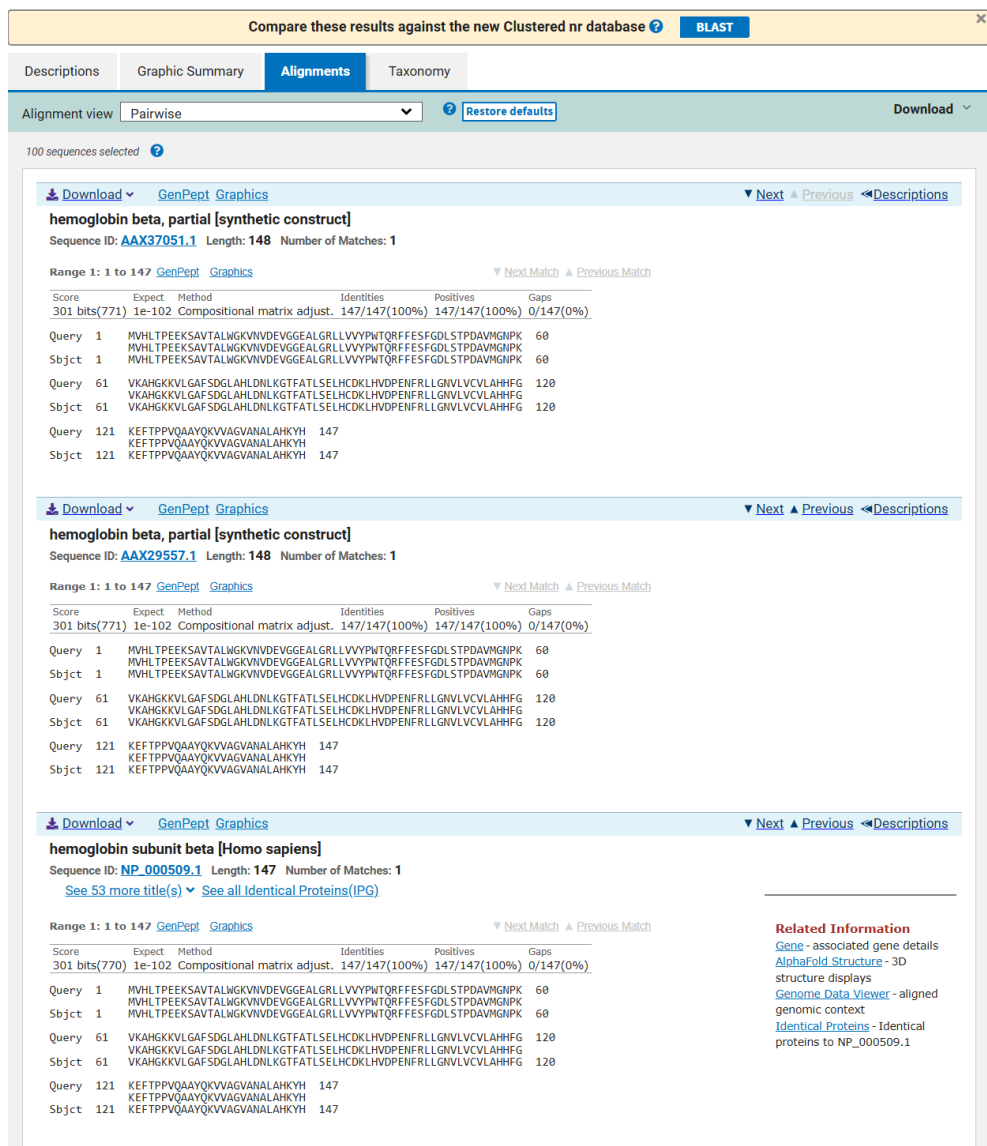


Figure 5: Pairwise alignment of the query sequence with human hemoglobin subunit beta.

Synthetic Construct Matches

Some top BLAST hits corresponded to synthetic β -globin constructs with 100% identity. Since these represent constructed sequences, the natural human reference sequence NP_000509.1 was selected as the primary biological reference

Homology with a Closely Related Species

A biologically relevant homolog identified in the BLAST results was hemoglobin subunit beta from Gorilla (XP_018891709.1). The gorilla sequence was 147 amino acids long and showed 100% query coverage, 146/147 (99%) identity, 147/147 (100%) positives, no gaps, and an E-value of 5e-102. Only one amino-acid position differed from the human sequence.

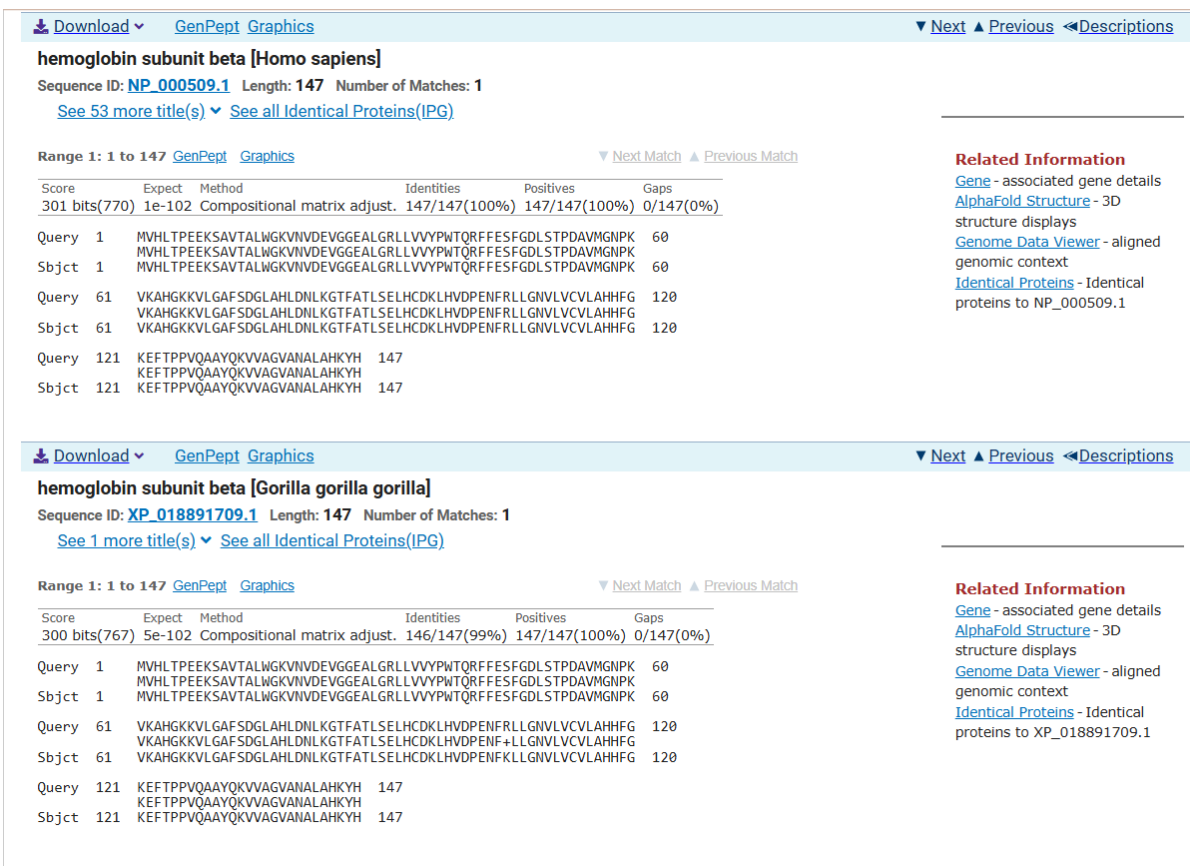


Figure 6. Sequence comparison and alignment of human and gorilla hemoglobin beta protein sequences showing conserved and variable amino-acid positions.

The comparison showed:

Parameter	Human β -globin	Gorilla β -globin
Accession	NP_000509.1	XP_018891709.1
Sequence length	147 aa	147 aa
Query coverage	100%	100%
Identity	147/147 (100%)	146/147 (99%)
Positives	147/147 (100%)	147/147 (100%)
Gaps	0/147 (0%)	0/147 (0%)
E-value	1e-102	5e-102

The human sequence showed complete identity with the query sequence, whereas the gorilla sequence differed at only one amino-acid position. This demonstrates the high degree of conservation of the hemoglobin beta protein between humans and gorillas.

Other top BLAST hits also included human beta-globin variants, showing very high sequence similarity.

Interpretation of BLAST Parameters

Percentage Identity

Percentage identity represents the proportion of amino acids that are exactly the same between the query and subject sequences.

The query showed **100% identity (147/147)** with human hemoglobin beta. This indicates that every amino acid in the query sequence matched the corresponding amino acid in the reference sequence.

Query Coverage

Query coverage indicates how much of the query sequence is included in the alignment.

The query coverage was **100%**, meaning that the complete 147-amino-acid sequence was aligned with the reference protein.

E-value

The E-value indicates the probability of obtaining a match of similar quality by chance in the database.

The E-value of **1e-102** is extremely small, indicating that the observed similarity is highly statistically significant and extremely unlikely to have occurred by chance.

Bit Score

The bit score represents the quality of the sequence alignment. A higher score indicates a better alignment.

The human hemoglobin beta match had a bit score of **301 bits**, indicating an exceptionally strong sequence match.

Gaps

The alignment contained **0 gaps**, meaning that no insertions or deletions were required to align the query with the human β -globin sequence.

Biological Significance

Hemoglobin is an important protein found in red blood cells and is responsible for transporting oxygen throughout the body. Adult human hemoglobin A consists primarily of two alpha and two beta subunits ($\alpha_2\beta_2$).

The beta subunit is encoded by the **HBB gene**. The BLASTP analysis identified the query sequence as human hemoglobin beta with complete sequence identity.

The strong similarity observed with hemoglobin beta proteins from other primates also demonstrates that this protein is highly conserved. The 99% identity observed with the gorilla sequence indicates that the protein sequence has remained very similar during the evolutionary divergence of humans and gorillas.

The presence of a highly significant E-value, complete query coverage, and 100% sequence identity provides strong evidence for the identification of the query protein as human hemoglobin beta.

Conclusion

The 147-amino-acid protein sequence was successfully analyzed using the NCBI BLASTP tool against the nr protein database. The analysis identified **hemoglobin subunit beta [Homo sapiens]**, accession **NP_000509.1**, as the best matching sequence.

The query sequence showed **147/147 (100%) amino-acid identity, 100% query coverage, 0 gaps**, an **E-value of 1e-102**, and a **bit score of 301 bits**. These results indicate an extremely strong and statistically significant match.

The comparison with the gorilla hemoglobin beta sequence also showed **99% identity**, demonstrating the evolutionary conservation of the beta-globin protein among closely related primates.

Therefore, based on the BLASTP results, the analyzed protein sequence can be confidently identified as **human hemoglobin subunit beta (β -globin)**.
